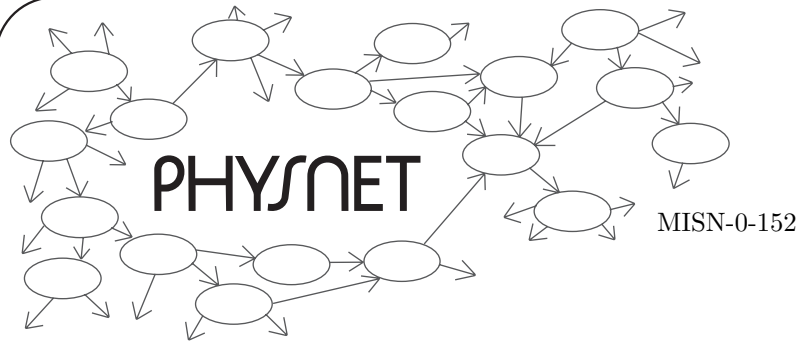
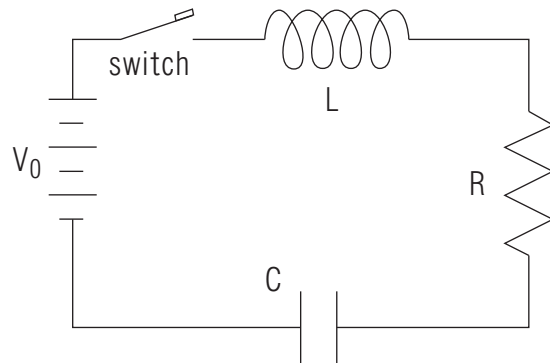




THREE-ELEMENT DC-DRIVEN SERIES LRC CIRCUIT



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by

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Input Skills:

1. Skills from “Two-Element D.C.-Driven LRC Circuits” (MISN-0-151).

Output Skills (Knowledge):

- K1. Starting from the charge-current and voltage-current relations for the three types of passive circuit element:
 - a. Derive the relation between the time rate of change of charge and the circuit parameters.
 - b. Given a solution for the relation, evaluate as many constants as possible without using any information about the circuit’s initial state.
 - c. Explain why two solution forms are necessary for the relation.

Post-Options:

1. “The Driven LRC Circuit: Resonances” (MISN-0-154).

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1. Introduction and Description

In this module we analyze the special case of the D.C.-driven series *LRC* circuit. By “series *LRC*” we mean there is an inductor, a resistor, and a capacitor connected in series. By “D.C.-driven” we mean that a direct-current voltage is applied for some period of time and there is no other source of EMF (see Fig. 1.). We examine the mathematics used to find a complete solution for the time dependence of the circulating charge.

2. Study Comments

The general method of attack on the D.C.-driven three-element circuit is the same as that used in the two-element case.¹ The reason the 3-element circuit is treated separately is that there are two solutions. To see why, we work through the circuit shown in Fig. 1.

Following the same procedure used for the 2-element circuit, we easily arrive at the fundamental equation for this circuit:

$$V_\epsilon = L \frac{d^2 q(t)}{dt^2} + R \frac{dq(t)}{dt} + \frac{1}{C} q(t) \quad (1)$$

A solution of Eq. (1) is:

$$q(t) = q_0 + q_1 e^{-at} \sin(bt + d), \quad (2)$$

¹See MISN-0-151.

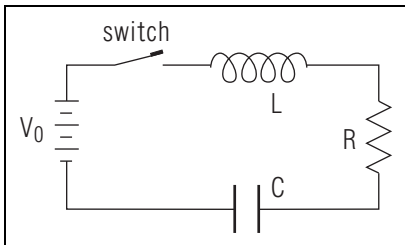


Figure 1. Switch open for $t < 0$, closed for $t > 0$.

where q_0 , q_1 , a , b and d are constants.

To prove that this is indeed a solution, we insert it and its first and second derivatives into Eq. (1). Collecting terms, we get:

$$V_\epsilon = \Lambda q_1 e^{-at} \sin(bt + d) + (Rb - 2Lab) q_1 e^{-at} \cos(bt + d) + \frac{q_0}{C}, \quad (3)$$

where we have defined for convenience:

$$\Lambda \equiv La^2 - Lb^2 - Ra + \frac{1}{C}.$$

Since Eq. (3) holds for all times t , we can evaluate the constants at any time we wish. To make our job easy we choose t such that $e^{-at} = 0$ ($t \rightarrow \infty$). We get:

$$V_\epsilon = \frac{q_0}{C},$$

so

$$q_0 = CV_\epsilon. \quad (4)$$

Putting this back into Eq. (3) and rearranging gives:

$$V_\epsilon - \Lambda q_1 e^{-at} \sin(bt + d) = V_\epsilon + (Rb - 2Lab) q_1 e^{-at} \cos(bt + d). \quad (5)$$

Cancelling terms,

$$-\Lambda \sin(bt + d) = (Rb - 2Lab) \cos(bt + d). \quad (6)$$

We now choose to evaluate Eq. (6) at $t = -d/b$ so $\sin(bt + d) = 0$ and $\cos(bt + d) = 1$, resulting in:

$$Rb - 2Lab = 0,$$

so:

$$a = \frac{R}{2L}. \quad (7)$$

After substituting this into Eq. (6), we pick t so that $\sin(bt + d) = 1$ and $\cos(bt + d) = 0$, giving:

$$Lb^2 - \frac{R^2}{4L} + \frac{R^2}{2L} - \frac{1}{C} = 0.$$

Solving for b gives:

$$b = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}. \quad (8)$$

Here is what we have found: Eq. (2) is a solution to Eq. (1) providing q_0 , a , and b are restricted to certain combinations of the circuit parameters. However, Eq. (2) is a solution to Eq. (1) no matter what the values of q_1 and d , so these latter are the two adjustable constants required in a solution of a second order differential equation. They must be set from the initial conditions for a particular problem you are wanting to solve.

The solution we have developed has a problem if $R^2/4L^2 > 1/LC$, for then the argument of the square root in Eq. (8) is negative! Then we must add a restriction to the solution we have found:

$$q(t) = CV_\epsilon + q_1 e^{-(R/2L)t} \sin \left(\sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} t + d \right); \quad R \leq 2\sqrt{L/C}. \quad (9)$$

Now what about cases that violate the restriction? For such cases we replace the circular functions, sine and cosine, with their hyperbolic counterparts:

$$\sinh(t) \equiv \frac{e^t - e^{-t}}{2} \quad \cosh(t) \equiv \frac{e^t + e^{-t}}{2}.$$

Note that the derivative of ($\cosh$) is ($+\sinh$), as opposed to the derivative of ($\cos$) being ($-\sin$) for the circular functions. We write this solution as:

$$q(t) = q_0 + q_1 e^{-at} \sinh(b't + d'),$$

where q_0 , q_1 , a , b' and d' are constants. Substitution into Eq. (1) produces:

$$V_\epsilon = \Lambda' q_1 e^{-at} \sinh(b't + d') + (Rb' - 2Lab') q_1 e^{-at} \cosh(b't + d') + \frac{q_0}{C}. \quad (10)$$

where:

$$\Lambda' \equiv La^2 + Lb'^2 - Ra + \frac{1}{C}.$$

Substituting into Eq. (10) and cancelling yields:

$$\left(-Lb'^2 - \frac{R^2}{4L} + \frac{R^2}{2L} - \frac{1}{C} \right) \sinh(b't + d') = 0.$$

Since this holds for any t , and $\sinh(b't + d')$ is not always zero, the coefficient of $\sinh(b't + d')$ must be zero:

$$-Lb'^2 - \frac{R^2}{4L} + \frac{R^2}{2L} - \frac{1}{C} = 0.$$

Thus:

$$b' = \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}},$$

which makes our second solution:

$$q(t) = CV_\epsilon + q_1 e^{-(R/2L)t} \sinh \left(\sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}} t + d' \right); \quad R \geq \sqrt{L/C}.$$

The unevaluated constants in our two solutions, q_1 and either d or d' , cannot be determined without specific information about the circuit's initial state.

Acknowledgments

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MODEL EXAM

$$q(t) = q_0 + q_1 e^{-at} \sin(bt + d)$$

$$q(t) = q_0 + q_1 e^{-at} \sinh(b't + d')$$

1. See Output Skill K1 in this module's *ID Sheet*.

Brief Answers:

1. See this module's *text*.

